above 1028 Kelvin, the three nongravitational forces appeared as one, as symmetric as they could possibly be. (At the end of this chapter we will discuss string theory's inclusion of the gravitational force into this high-temperature merger.) But as the temperature dropped below 1028 Kelvin, the universe underwent a phase transition in which the three forces crystallized out from their common union in different ways. Their relative strengths and the details of how they act on matter began to diverge. And so, the symmetry among the forces evident at higher temperatures was broken as the universe cooled. Nevertheless, the work of Glashow, Salam, and Weinberg (see Chapter 5) shows that not all of the high-temperature symmetry was erased: The weak and electromagnetic forces were still deeply interwoven. As the universe further expanded and cooled, nothing much happened until things simmered down to 1015 Kelvin—about 100 million times the sun's core temperature—when the universe went through another phase transition that affected the electromagnetic and weak forces. At this temperature, they too crystallized out from their previous, more symmetric union, and as the universe continued to cool, their differences became magnified. The two phase transitions are responsible for the three apparently distinct nongravitational forces at work in the world, even though this review of cosmic history shows that the forces, in fact, are deeply related.

A Cosmological Puzzle

This post-Planck era cosmology provides an elegant, consistent, and calculationally tractable framework for understanding the universe as far back as the briefest moments after the bang. But, as with most successful theories, our new insights raise yet more detailed questions. And it turns out that some of these questions, while not invalidating the standard cosmological scenario as presented, do highlight awkward aspects that point toward the need for a deeper theory. Let's focus on one. It is called the horizon problem, and it is one of the most important issues in modern cosmology.

Detailed studies of the cosmic background radiation have shown that regardless of which direction in the sky one points the measuring antenna, the temperature of the radiation is the same, to about one part in 100,000. If you think about it for a moment, you will realize that this is quite strange. Why should different locations in the universe,